DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF COMPUTING

UNIVERSITI TEKNOLOGI MALAYSIA

SECP3843 - SPECIAL TOPIC IN DATA ENGINEERING

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Tutorial 3 : AI Algorithm CNN

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What is Image Classification?

Image classification is supervised machine learning where an algorithm assigns a single class label to an input image based on its visual content. Mathematically, given an 60,000 image and a set of 10 possible classes including airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The model learns a mapping to output the class with the highest probability

What is CNN?

A Convolutional Neural Network is a deep learning architecture designed for grid-structured data like 2D images. CNN optimize the visual processing via local receptive fields (localized analysis), shared weights (consistent filter application), and hierarchical feature composition (building complex shapes from simple edges).

Core Architecture Layer

- Convolutional Layer (Conv2D)
  Slides learnable filters across the input to generate feature maps. Early layers detect basic edges and deeper layers for complex object parts.

- Activation Function (ReLU)
  Replaces negative values with zero to introduce non-linearity enabling the network to learn complex and real world patterns.

- Pooling Layer (MaxPooling2D)
  Down-samples feature maps by extracting the maximum value within a moving window for example 2 x 2 pixels. This reduces the data size, complexity and grant a degree of translation invariance

- Dropout Layer
  Randomly deactivates a fraction of neurons during training. This acts as a regularizer to prevent overfitting

- Flatten
  Converts multi-dimensional feature maps into a 1D vector, preparing the data for the final classification layers

- Dense Layer and Softmax:
  Connect all remaining features to calculate final class scores. The terminal layer uses Softmax to convert the scores into probabilities distribution over the 10 target classes.

Evaluation of CNN

To quantify a trained CNN's performance on unseen data, the following evaluation metrics and tools are utilized :

- Accuracy
  The proportion of correctly classified samples out of the total train dataset. It can show the performance weaknesses if the dataset has class imbalance.

- Loss
  The cost function is minimized by the optimizer during training. Lower value indicates better performance. Observing the gap between training and validation loss helps identify overfitting

- Confusion Matrix
  A 10 x 10 table mapping true classes against predicted classes. The diagonal cell represent correct classification while off diagonal cells highlight the specific misclassifications between classes.

- Training Curves
  Plotting accuracy and loss against epoch number reveal

Steps hand on in Python

```
(x_train, y_train), (x_test, y_test) = datasets.cifar10.load_data()

class_names = ['airplane', 'automobile', 'bird', 'cat', 'deer',
               'dog', 'frog', 'horse', 'ship', 'truck']

print('Train images shape:', x_train.shape)
print('Train labels shape:', y_train.shape)
print('Test images shape :', x_test.shape)
print('Test labels shape :', y_test.shape)
print('Pixel value range :', x_train.min(), '-', x_train.max())
```

```
x_train = x_train.astype('float32') / 255.0
x_test  = x_test.astype('float32')  / 255.0

y_train_cat = to_categorical(y_train, 10)
y_test_cat  = to_categorical(y_test, 10)

print('After normalization, pixel range:', x_train.min(), '-', x_train.max())
print('One-hot label shape (train):', y_train_cat.shape)
print('Example one-hot label:', y_train_cat[0])
```

```
model = models.Sequential([
    layers.Input(shape=(32, 32, 3)),

    layers.Conv2D(32, (3, 3), activation='relu', padding='same'),
    layers.Conv2D(32, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),

    layers.Conv2D(64, (3, 3), activation='relu', padding='same'),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),

    layers.Flatten(),
    layers.Dense(512, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(10, activation='softmax'),
])

model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy'],
)

model.summary()
```

```
EPOCHS = 10
BATCH = 64

history = model.fit(
    x_train, y_train_cat,
    epochs=EPOCHS,
    batch_size=BATCH,
    validation_data=(x_test, y_test_cat),
    verbose=1,
)
```

What is the weakest in the current CNN model?

How to enhance the CNN model.

What is the evaluation based on CNN and the new_CNN model?

Results and Discussion

Reflection

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